SP25 CSE 5914
Knowledge Based Systems

Group: The BEST

Timebox 1

Our Crew

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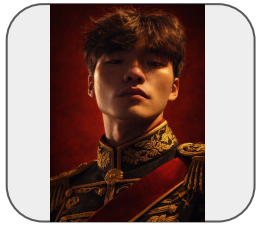

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Idea: FITT

(Fitness/Nutrition/Wellness App)

Description:

- A **gym-centric AI system** that optimizes strength and hypertrophy by continuously analyzing workout performance.
- Tracks **sets, reps, weight, RPE, rest times, and progression trends** as the primary signal.
- Uses secondary signals (sleep, nutrition, recovery, stress) **only to explain or correct performance issues**.
- Connects to health app
- Automatically adjusts:
    - Training volume (deloads, overloads)
    - Rep schemes and weight progression
    - Caloric intake and macronutrients
    - Sleep and recovery recommendations

- Learns the user's individual response patterns over time using a trained model.

Similar ideas/applications on market:

- Strong
- Hevy
- Fitbod
- RP Hypertrophy
- MyFitnessPal
- WHOOP
- Garmin

Most tools track data **in isolation,** while requiring **manual interpretation** and does **not connect gym performance to recovery decisions.**

How does our idea differentiate

1. **Performance-aware deloading:** Deloads are triggered by real fatigue signals, not fixed schedules.

2. **Adaptive nutrition:** Calories and macros change only when performance demands it.

3. **Gym-first decision making:** Training performance is the ground truth, not calories or sleep alone.

4. **Personalized model:** Learns how *your body* responds to volume, intensity, sleep, and food over time.

5. **Closed-loop AI system:** Observe → Diagnose → Adjust → Re-evaluate automatically.

6. **Explainable AI:** Every recommendation includes a rationale tied to performance data.

Persona/s & Use Cases

Dylan
Age: 25
Occupation: Software Engineer
Location: St. Louis, Missouri

Gym Habits:

- Uses a standard workout app like Strong.
- Performs the exact same workout with the exact same weights.
- Wears an apple watch to track his sleep, but never looks at how a bad night of sleep affects his strength the next day.

Pain points:

- Dylan has been lifting the same weight on bench press for 3 months. He is frustrated because his current app doesn't tell him how to break through this plateau.
- Sometimes he doesn't know if he's actually tired or not. He lacks a system that says "you are fatigued, take the day off."
- Information overload when he sees influencers talk about "macros" or "progressive overload" on TikTok, and has no idea how to apply those math-heavy concepts to himself.

Use case we'll support:

- Instead of guessing, Dylan can use FITT to see all his reps easily and say "Your performance was high-velocity today. Next week, we are increasing your weight by 5 lbs," solving the plateau.
- If Dylan's strength drops 10%, FITT will look at his sleep data and say "Your recovery is low. Today's workout is reduced by 3 sets to prevent injury." Hence, no guesswork.
- Adaptive nutrition sync for when if Dylan has a high-intensity leg day, FITT automatically tells him "You burned extra energy today. Increase your protein by 20g to help your muscles repair."

Technology Stack & Data Set

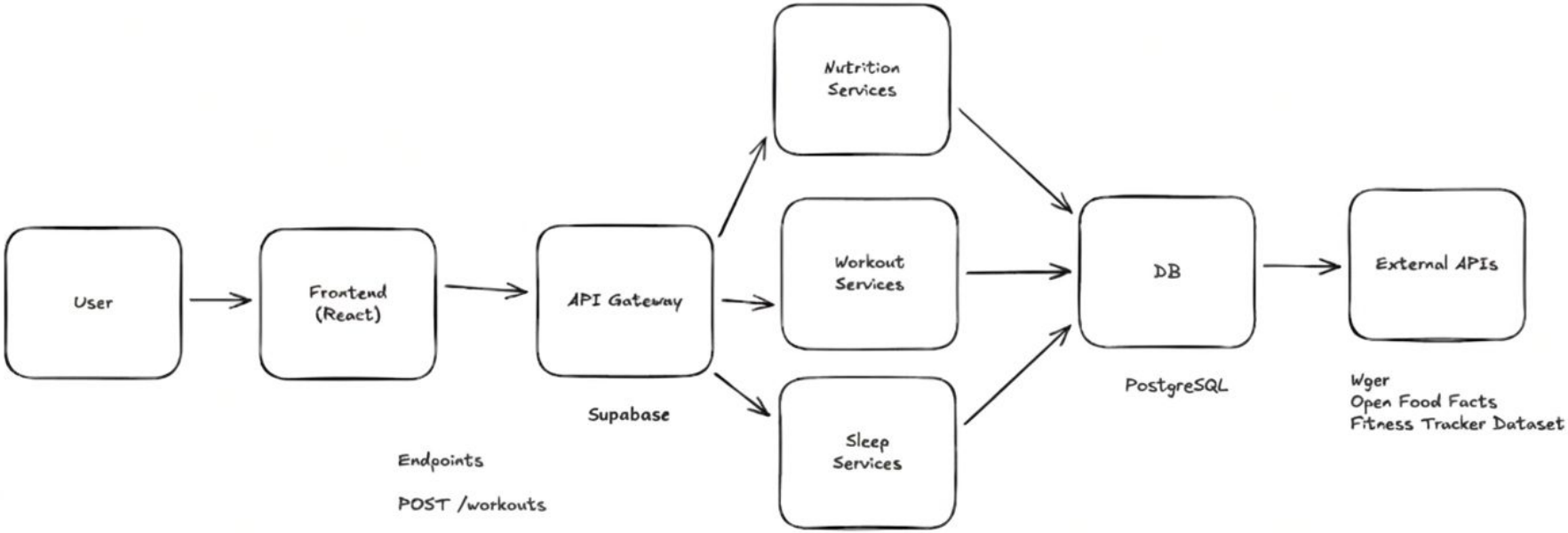
Technologies:

- **Supabase** – for handling user requests and serve exercise and nutrition data.
- **PostgreSQL** – for storing user profiles, workout logs, and meal records.
- **Wger API** – for providing structured exercise data including names, muscle groups, equipment, and instructions.
- **Open Food Facts API** – for retrieving nutrition information for foods by barcode or search terms.
- **Pandas** – for processing and analyzing CSV datasets like the Kaggle Fitness Tracker data.
- **Groq AI** – for interpreting natural language queries and generating conversational responses.
- **React** – for building a web or mobile frontend interface for the user.

APIs:

- **Wger API**
  - The Wger API provides a comprehensive exercise database with structured information about exercises, muscles, equipment, categories, and images.
  - Its purpose is to provide the exercises that FITT will choose from
- **Open Food Facts API**
  - Provides accurate nutrition information for foods, enabling calorie and macronutrient tracking.
  - Its purpose is to provide the food data and calories for the meal tracker portion of FITT
- **Fitness Tracker Dataset**
  - The Fitness Tracker Dataset from Kaggle contains daily activity records including BPM both average and max, workout type, workout duration, and BMI, and we are using it to track user progress, personalize workout and nutrition recommendations, and analyze trends over time.

User → Frontend → API Gateway → Services → Datastores → External APIs



Endpoints

POST /workouts

GET /workouts/{workout_id}

DELETE /workouts/{workout_id}

Exercise catalog

GET /exercises/search?q=

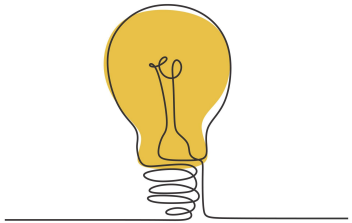
GET /exercises/{exercise_id}

Recommendations

- How to use user data: Using data from the Kaggle Fitness Tracker Dataset - includes age, gender, weight, height, BPM, etc.

- Connects to health app

- Set up git and cloud accounts for Timebox 2



Roadmap

01

- Formed team, selected FITT idea
- Identified how to use user data
- Defined tech stacks and APIs
- Created/shaped persona
- Sample application design
- Set up project plan & preliminary structure.

02

- Set up cloud and git environment
- Finalize API connections
- Build initial database schema in PostgreSQL
- Begin working on the different services
- Begin to get basic function of app

03

- Fix any bugs from Timebox 2
- Finish Frontend
- Begin Real World Testing

04

- Fix any bugs from Timebox 3
- Finalize any new/recommended features
- Prepare final demo.

Other Info

Kaggle Data Set - https://www.kaggle.com/datasets/nadeemajeedch/fitness-tracker-dataset/data

Wger API - https://wger.readthedocs.io/en/latest/

Open Food Facts API - https://world.openfoodfacts.org/